CS24305 • MODULE 2

Module 2: Data Encoding & Multiplexing Techniques

Topics 14 to 19 • Line Coding, Scrambling, QAM Modulation, PCM & TDM/FDM

Module II: Transmission Media & Signal Encoding — Complete 6-Topic Tracker

Topic 14: Guided Media (Twisted Pair UTP/STP, Coaxial Cable, Optical Fiber)

Topic 15: Wireless Media & Propagation (Ground, Sky, LOS, Satellite Microwave)

Topic 16: Digital Signaling (Line Codes: NRZ-L, NRZ-I, Manchester, AMI)

Topic 17: Analog Signaling (Carrier Parameter Representations)

Topic 18: Encoding Techniques (Line Coding Comparison & Tradeoffs)

Topic 19: Modulation Techniques (ASK, FSK, PSK & Constellation QAM)

Topic 14: Guided Transmission Media

Guided Medium	Physical Architecture & Transmission Physics	Key Advantages & Tradeoffs
1. Twisted Pair Cable	Two insulated copper wires twisted in a helical spiral. Twisting ensures both wires receive equal electromagnetic interference (EMI), which cancels out at the differential receiver. Available as **UTP** (Unshielded) and **STP** (Shielded with metal foil).	Inexpensive, lightweight, easy to terminate with RJ-45 connectors. Limited bandwidth ($\leq 10\text{ Gbps}$ over short distances), prone to crosstalk.
2. Coaxial Cable	Solid central copper conductor surrounded by dielectric insulating layer, enclosed in a braided cylindrical wire mesh shield and outer protective plastic jacket.	Better noise immunity and higher bandwidth than twisted pair. Bulkier, stiffer, harder to install. Used in Cable TV broadband (DOCSIS).
3. Optical Fiber	Ultra-pure cylindrical glass core (n_1) surrounded by concentric glass cladding (n_2) where refractive index $n_1 > n_2$. Transmits light pulses via **Total Internal Reflection** whenever incident angle $\theta > \theta_{\text{critical}}$.	Enormous bandwidth (terabits/sec), ultra-low attenuation ($\approx 0.2\text{ dB/km}$), zero EMI susceptibility, lightweight, highly secure against wiretapping. Expensive splicing and optical transceivers.

Topic 15: Wireless Transmission & Propagation [UPLOADED PYQ]

Propagation Mode	Atmospheric Interaction & Frequency Band	Representative Applications
1. Ground-Wave Propagation	Signals follow the curvature of the earth via diffraction over the conductive ground surface ($< 2\text{ MHz}$).	AM radio broadcasting, maritime navigation.
2. Sky-Wave Propagation	Signals bounce back and forth between earth's surface and the charged ionosphere ($2\text{–}30\text{ MHz}$).	Shortwave (SW) international radio broadcasting, amateur radio.
3. Line-of-Sight (LOS)	Straight-line path between transmitting and receiving directional horn antennas ($> 30\text{ MHz}$). Antennas must have clear Fresnel zone.	Terrestrial microwave links, cellular base stations, Wi-Fi.

Satellite Microwave vs. Terrestrial Microwave [UPLOADED PYQ]:

- **Terrestrial Microwave:** Uses parabolic dish antennas mounted on high towers spaced $30\text{–}50\text{ km}$ apart due to Earth's horizon curvature. Low propagation delay ($< 1\text{ ms}$).
- **Satellite Microwave:** Earth station beams signal to a geostationary satellite at $35,786\text{ km}$ altitude (Uplink), which amplifies and retransmits it on a different downlink frequency. High round-trip latency ($\approx 270\text{–}540\text{ ms}$), but provides massive continental coverage.

Topic 16 & 18: Digital Line Coding Schemes [UPLOADED PYQ]

[UPLOADED PYQ] Step-by-Step Waveform Rules for Bit Stream `001110011`

Encoding Scheme	Bit Transition Invariant Rule	Waveform Encoding for `001110011`
NRZ-L (Level)	Bit `0` = Positive voltage ($+V$), Bit `1` = Negative voltage ($-V$).	`+V, +V, -V, -V, -V, +V, +V, -V, -V`
NRZ-I (Invert)	Bit `1` = Transition at start of bit interval; Bit `0` = No transition.	Maintains previous level on `0`, flips on `1`.
Bipolar AMI	Bit `0` = Zero voltage ($0V$); Bit `1` = Alternating $+V$ and $-V$.	`0V, 0V, +V, -V, +V, 0V, 0V, -V, +V`
Manchester (802.3)	Bit `0` = High-to-Low transition at bit center; Bit `1` = Low-to-High transition at center.	Guarantees a mid-bit clock transition in every single bit!
Differential Manchester	Always transitions at center. Bit `0` = Transition at start; Bit `1` = No transition at start.	Token Ring standard; immune to wire polarity inversion.

Topic 17 & 19: Carrier Modulation Techniques (ASK, FSK, PSK, QAM)

Digital modulation alters one or more parameters of a high-frequency analog sinusoidal carrier $c(t) = A\cos(2\pi f_c t + \phi)$:

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Modulation	Modulated Parameter	Mathematical Signal Equation	Key Advantages & Tradeoffs
ASK	Amplitude	$s(t) = A_1 \cos(2\pi f_c t)$ for `1`, $A_0 \cos(2\pi f_c t)$ for `0`	Simple; highly susceptible to noise and fading.
FSK	Frequency	$s(t) = A \cos(2\pi f_1 t)$ for `1`, $A \cos(2\pi f_2 t)$ for `0`	High noise immunity; requires more channel bandwidth.
BPSK / QPSK	Phase	$s(t) = A \cos(2\pi f_c t + \phi_i)$ where $\phi_i \in \{0, \pi\}$	Excellent bit error rate (BER); standard in modern wireless.
QAM (16/64/256)	Amplitude + Phase	$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t)$	Enormous spectral efficiency ($\log_2 M$ bits/baud); used in Wi-Fi & 5G.

🧠 M2 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Draw the line coding waveforms for bit stream `001110011` using NRZ-L, NRZ-I, Bipolar AMI, Manchester, and Differential Manchester. (10 Marks)

1. **NRZ-L:** High for `0`, Low for `1` $\rightarrow$ `High, High, Low, Low, Low, High, High, Low, Low`.
2. **NRZ-I:** Invert signal level on every `1`; no change on `0`.
3. **Bipolar AMI:** Zero volts for all `0`s; alternating $+V$ and $-V$ for each successive `1` $\rightarrow$ `0, 0, +V, -V, +V, 0, 0, -V, +V`.
4. **Manchester:** Every bit has mid-interval transition. `0` = High-to-Low; `1` = Low-to-High.
5. **Differential Manchester:** Mandatory mid-bit transition. Start-of-bit transition for `0`; no start transition for `1`.

Topic 19.2: Advanced Modulation, Constellation Geometry & Delta Modulation

In modern high-speed modems (such as V.90, V.92, DSL, DOCSIS cable modems, and 4G/5G LTE), multi-dimensional constellation geometry allows packing multiple bits per symbol over analog carrier channels.

🏛 Step-by-Step Delta Modulation (DM) & Slope Overload Distortion

Delta Modulation (DM) transmits only 1 bit per sample, indicating whether the current analog sample amplitude is higher (Bit 1: $+\delta$) or lower (Bit 0: $-\delta$) than the previous staircase approximation.

- **Slope Overload Distortion:** Occurs when the input analog signal changes faster than the maximum rate of staircase increase ($|\frac{dx(t)}{dt}| > \frac{\delta}{T_s}$). The staircase cannot keep up with rapid signal rises.
- **Granular Noise:** Occurs when the input analog signal is flat or slowly varying. The staircase oscillates continuously above and below the flat signal by $\pm\delta$.
- **Adaptive Delta Modulation (ADM):** Dynamically scales step size δ based on successive bit history (doubles δ on consecutive 1s to prevent slope overload; shrinks δ on alternating 0101 to eliminate granular noise).

🏛 Solved Numerical: T1 and E1 Digital Carrier Frame Anatomy

1. North American T1 Carrier System (24 Voice Channels):

$$\text{Each voice channel} = 8 \text{ bits/sample} \times 8000 \text{ samples/sec} = 64 \text{ kbps}$$

$$\text{T1 Frame} = (24 \text{ channels} \times 8 \text{ bits}) + 1 \text{ Framing Bit} = 192 + 1 = \mathbf{193} \text{ bits/frame}$$

$$\text{T1 Bit Rate} = \mathbf{193} \text{ bits/frame} \times \mathbf{8000} \text{ frames/sec} = \mathbf{1,544,000} \text{ bps (1.544 Mbps)}$$

2. European E1 Carrier System (32 Voice Channels):

$$\text{E1 Frame} = 32 \text{ channels} \times 8 \text{ bits} = \mathbf{256} \text{ bits/frame}$$

$$\text{E1 Bit Rate} = \mathbf{256} \text{ bits/frame} \times \mathbf{8000} \text{ frames/sec} = \mathbf{2,048,000} \text{ bps (2.048 Mbps)}$$

Note: Channel 0 is reserved for framing/synchronization; Channel 16 is reserved for out-of-band signaling (SS7).

Topic 19.3: Comprehensive Modulation Geometry & Companding Physics

In Pulse Code Modulation (PCM), quantization of speech signals causes higher relative distortion for soft whisper sounds than for loud vowel sounds when linear uniform quantization intervals are used. Non-linear **Companding (Compressor-Expander)** restores a constant dynamic Signal-to-Quantization-Noise Ratio across all volume levels.

The 2 Global Companding Laws:
1. North American μ-Law Companding ($\mu = 255$):

$$\mathbf{y} = \operatorname{sgn}(\mathbf{x}) \frac{\ln(\mathbf{1} + \mu|\mathbf{x}|)}{\ln(\mathbf{1} + \mu)} \quad \text{for } -\mathbf{1} \le \mathbf{x} \le \mathbf{1}$$

2. European A-Law Companding ($A = 87.6$):

$$\mathbf{y} = \operatorname{sgn}(\mathbf{x}) \begin{cases} \frac{A|x|}{1+\ln(A)} & 0 \le |x| \le \frac{1}{A} \\ \frac{1+\ln(A|x|)}{1+\ln(A)} & \frac{1}{A} \le |x| \le 1 \end{cases}$$

🏛 Step-by-Step Solved Problem: Detailed Constellation Diagram Bit Mapping for QPSK & 16-QAM

In Gray-coded 16-QAM, each 4-bit symbol $(b_1 b_2 b_3 b_4)$ is mapped into two independent in-phase (I) and quadrature (Q) amplitudes taking values from $\{-3d, -d, +d, +3d\}$:

Symbol ($b_1 b_2$)	In-Phase (I) Axis	Symbol ($b_3 b_4$)	Quadrature (Q) Axis	Net Euclidean Distance
`00`	$-3d$	`00`	$+3d$	$d_{\min} = 2d$
`01`	$-1d$	`01`	$+1d$	$d_{\min} = 2d$
`11`	$+1d$	`11`	$-1d$	$d_{\min} = 2d$
`10`	$+3d$	`10`	$-3d$	$d_{\min} = 2d$

Gray Coding Property: Adjacent constellation points differ by strictly **one bit**. Thus, the most probable symbol errors (noise nudging a point into an adjacent decision zone) cause only a single-bit error!

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🏛️ Solved Numerical: SONET / SDH Optical Multiplexing Hierarchy

The Synchronous Optical Network (SONET) base electrical signal is **STS-1** (Synchronous Transport Signal level 1), corresponding to optical carrier **OC-1**.

$$\text{STS-1 Frame Layout} = 9 \text{ rows} \times 90 \text{ columns (bytes)} = \mathbf{810} \text{ bytes/frame}$$

$$\text{Frame Rate} = 8000 \text{ frames/sec (Frame duration } = 125\ \mu\text{s)}$$

$$\text{STS-1 Data Rate} = \mathbf{810} \text{ bytes/frame} \times \mathbf{8} \text{ bits/byte} \times \mathbf{8000} \text{ frames/sec} = \mathbf{51.84} \text{ Mbps}$$

- $\text{OC-3} = 3 \times 51.84 = \mathbf{155.52}$ Mbps (Matches European STM-1 standard)
- $\text{OC-12} = 12 \times 51.84 = \mathbf{622.08}$ Mbps (Matches European STM-4 standard)
- $\text{OC-48} = 48 \times 51.84 = \mathbf{2.488}$ Gbps (Matches European STM-16 standard)
- $\text{OC-192} = 192 \times 51.84 = \mathbf{9.953}\text{ Gbps} \approx \mathbf{10}\text{ Gbps}$

Topic 19.4: Master University Solved Examination Numerical Bank

🏛️ Solved Numerical 2: Line Coding Baud Rate & Bandwidth Requirements

A digital bitstream of rate $R = 10$ Mbps is transmitted using: (a) NRZ-L, (b) Manchester, (c) 4B/5B Block Coding with NRZ-I. Calculate the signal baud rate (S) and minimum Nyquist bandwidth ($B_{\min}$) for each scheme.

Solution: Baud Rate $S = c \times R \times \frac{1}{r}, \quad B_{\min} = \frac{S}{2}$ (for average factor $c = 1/2$).

- **(a) NRZ-L ($r = 1$):** Baud rate $S = 10$ Mbaud $\implies B_{\min} = \frac{10}{2} = \mathbf{5}$ MHz.
- **(b) Manchester ($r = 0.5$):** Baud rate $S = 20$ Mbaud $\implies B_{\min} = \frac{20}{2} = \mathbf{10}$ MHz (Requires $2\times$ the bandwidth of NRZ!).
- **(c) 4B/5B with NRZ-I ($r = 4/5 = 0.8$):** Baud rate $S = \frac{10}{0.8} = 12.5$ Mbaud $\implies B_{\min} = \frac{12.5}{2} = \mathbf{6.25}$ MHz (Only 25% bandwidth overhead with guaranteed clock synchronization!).

🏛️ Solved Numerical 3: Pulse Code Modulation (PCM) of Video Signals

A television video signal has a maximum frequency bandwidth of $f_{\max} = 4.5$ MHz. The signal is sampled at 20% above the Nyquist rate and quantized into 1024 discrete levels ($L = 1024$). Calculate: (1) Sampling rate, (2) Bits per sample, (3) Output digital bit rate, and (4) Minimum channel bandwidth.

Solution:

$$1. \text{ Nyquist Rate } = 2 \times 4.5\text{ MHz} = 9\text{ MHz} \implies f_s = 1.20 \times 9\text{ MHz} = \mathbf{10.8}\text{ MHz (10.8 million samples/sec)}$$

$$2. \text{ Bits per Sample } n = \log_2(1024) = \mathbf{10} \text{ bits/sample}$$

$$3. \text{ Bit Rate } R = f_s \times n = 10.8\text{ MHz} \times 10 = \mathbf{108}\text{ Mbps}$$

$$4. \text{ Minimum Bandwidth } B_{\min} = \frac{R}{2} = \frac{108}{2} = \mathbf{54}\text{ MHz}$$

Topic 19.5: Mathematical Derivation of PCM Quantization Noise & Constellation Theory

In Pulse Code Modulation (PCM), a continuous analog signal $x(t)$ with peak-to-peak voltage range $[-V_{\max}, +V_{\max}]$ (total voltage span $2V_{\max}$) is divided into $L = 2^n$ uniform quantization intervals of step width:

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Quantization Step Size & Quantization Noise Power Derivation:

$$\mathbf{\Delta = \frac{2V_{\max}}{L} = \frac{2V_{\max}}{2^n}}$$

Assuming the quantization error $e = x - x_q$ is uniformly distributed over $\left[-\frac{\Delta}{2}, +\frac{\Delta}{2}\right]$ with probability density function $f_E(e) = \frac{1}{\Delta}$:

$$\sigma_q^2 = \int_{-\Delta/2}^{+\Delta/2} \mathbf{e^2 f_E(e)\, de} = \mathbf{\frac{1}{\Delta}} \left[\mathbf{\frac{e^3}{3}}\right]_{-\Delta/2}^{+\Delta/2} = \mathbf{\frac{1}{\Delta}} \left(\mathbf{\frac{\Delta^3}{24}} - \left(-\mathbf{\frac{\Delta^3}{24}}\right)\right) = \mathbf{\frac{\Delta^2}{12}}$$

$$\text{Signal-to-Quantization-Noise Ratio: } \mathrm{SQNR_{dB}} = \mathbf{10}\log_{\mathbf{10}}\left(\frac{\mathbf{P}_{\text{signal}}}{\sigma_{\mathbf{q}}^{\mathbf{2}}}\right) = \mathbf{6.02n + 1.76}\ \mathrm{dB}$$

Engineering Axiom: Every additional bit added to each PCM sample increases the Signal-to-Noise Ratio by exactly $\mathbf{6.02}$ dB!

🏛️ Complete Comparison Matrix: Digital Line Coding Spectral Properties

Coding Scheme	Baud Rate (S)	DC Component?	Synchronization Quality	Primary Real-World Standard
NRZ-L	$S = N$ baud	Severe DC bias	Poor (loses clock on 0s & 1s)	RS-232 Serial COM Ports
NRZ-I	$S = N$ baud	Severe DC bias	Partial (syncs on 1s only)	USB 1.1/2.0 (with Bit Stuffing)
Bipolar AMI	$S = N$ baud	**Zero DC Bias**	Partial (syncs on 1s only)	ISDN Primary Rate, DS1 lines
Manchester	$S = 2N$ baud	**Zero DC Bias**	**100% Guaranteed**	IEEE 802.3 10BASE-T Ethernet
Diff. Manchester	$S = 2N$ baud	**Zero DC Bias**	**100% Guaranteed**	IEEE 802.5 Token Ring LANs
4B/5B + NRZ-I	$S = 1.25N$ baud	Low DC bias	**100% Guaranteed**	100BASE-TX Fast Ethernet, FDDI
8B/10B Coding	$S = 1.25N$ baud	Zero DC bias	**100% Guaranteed**	Gigabit Ethernet (1000BASE-X), PCIe, SATA

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🏛 Solved Numerical 4: Wavelength-Division Multiplexing (WDM) Throughput

A single-mode fiber optical communication link utilizes Dense Wavelength Division Multiplexing (DWDM) operating in the C-band ($1530 \text{ nm to } 1565 \text{ nm}$). The system multiplexes 80 distinct optical wavelengths, each modulated at 100 Gbps using DP-QPSK. Calculate: (1) Total aggregate throughput of the single fiber strand in Tbps, and (2) Total bits transmitted in 1 hour.

Solution:

$$\text{Aggregate Data Rate } R = 80 \text{ channels} \times 100 \text{ Gbps/channel} = \mathbf{8000} \text{ Gbps} = \mathbf{8} \text{ Tbps (Terabits/sec)}$$

$$\text{Total Bits in 1 Hour } = 8 \times 10^{12} \text{ bps} \times 3600 \text{ s} = \mathbf{2.88} \times \mathbf{10^{16}} \text{ bits} = \mathbf{3.6} \text{ Petabytes (PB)}$$

Q6. Compare ASK, FSK, BPSK, QPSK, and 16-QAM across bandwidth efficiency, noise immunity, and constellation complexity. (10 Marks)

• **ASK:** Simple hardware; highly vulnerable to noise and attenuation; low spectral efficiency (1 bit/baud).
• **FSK:** High noise immunity; requires large bandwidth (two separate carrier frequencies); standard in low-speed modems (1 bit/baud).
• **BPSK:** Excellent noise immunity (opposite phases $\pm 180°$); 1 bit/baud.
• **QPSK:** High noise immunity with $2\times$ spectral efficiency (2 bits/baud); standard in satellite/cellular links.
• **16-QAM:** Combines amplitude and phase; high spectral efficiency (4 bits/baud); requires high SNR to distinguish adjacent points.

Q7. Explain Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM), and Pulse Position Modulation (PPM). (8 Marks)

These are analog pulse modulation techniques where a periodic pulse train carrier is modulated by an analog message:
1. **PAM:** The amplitude of each pulse is proportional to the instantaneous analog sample amplitude (intermediate step in PCM).
2. **PWM:** The width (duration) of each pulse varies proportionally with sample amplitude (used in motor control, audio amplifiers).
3. **PPM:** The relative position (timing offset) of each constant-width pulse varies proportionally with sample amplitude (high noise immunity).

Topic 19.6: Exhaustive Digital Carrier Modulation & Constellation Geometry

Digital passband transmission utilizes Orthogonal In-Phase (I) and Quadrature (Q) carrier bases to synthesize high-density signal constellations:

$$s(t) = I(t)\sqrt{\frac{2}{T_s}}\cos(2\pi f_c t) - Q(t)\sqrt{\frac{2}{T_s}}\sin(2\pi f_c t)$$

Modulation Scheme	Bits per Symbol (k)	Spectral Efficiency (bps/Hz)	Minimum SNR Required	Representative Technology
BPSK	1 bit	1.0 bps/Hz	Low (7 dB)	Deep Space, Bluetooth LE
QPSK	2 bits	2.0 bps/Hz	Moderate (10 dB)	DVB-S Satellite, 3G UMTS
8-PSK	3 bits	3.0 bps/Hz	Moderate (14 dB)	EDGE Cellular (2.75G)
16-QAM	4 bits	4.0 bps/Hz	High (17 dB)	Wi-Fi 4 (802.11n), 4G LTE
64-QAM	6 bits	6.0 bps/Hz	Very High (23 dB)	Wi-Fi 5 (802.11ac), DVB-C
256-QAM	8 bits	8.0 bps/Hz	Extremely High (29 dB)	Wi-Fi 6 (802.11ax), DOCSIS 3.1
1024-QAM	10 bits	10.0 bps/Hz	Ultra High (35 dB)	Wi-Fi 6E / Wi-Fi 7 (802.11be)
4096-QAM	12 bits	12.0 bps/Hz	Peak Clean SNR (41 dB)	Wi-Fi 7 Enterprise Channels

🏛 Solved Problem: Detailed Pulse Code Modulation (PCM) Dynamic Range & SNR Analysis

An analog instrumentation sensor signal has bandwidth $f_{\text{max}} = 10 \text{ kHz}$ and a required dynamic range of 72 dB.

1. What is the minimum number of PCM quantization bits (n) required to achieve $\text{SQNR} \geq 72 \text{ dB}$?
2. What is the resulting output digital bit rate when sampled at 25% above the Nyquist rate?

Solution:

$$\text{SQNR}_{\text{dB}} \approx 6.02n + 1.76 \geq 72 \text{ dB} \implies 6.02n \geq 72 - 1.76 = 70.24$$

$$n \geq \frac{70.24}{6.02} \approx 11.67 \implies \mathbf{n = 12} \text{ bits/sample} \quad (\mathbf{L = 2^{12} = 4096} \text{ levels})$$

$$\text{Sampling Rate: } f_s = 1.25 \times (2 \times 10 \text{ kHz}) = 1.25 \times 20 \text{ kHz} = \mathbf{25} \text{ kHz } (25{,}000 \text{ samples/sec})$$

$$\text{Output Bit Rate: } \mathbf{R = f_s \times n = 25,000 \times 12 = 300,000} \text{ bps } (300 \text{ kbps})$$

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Topic 19.7: Master University Long-Answer Exam Solutions Bank

Q8. Explain the 4B/5B and 8B/10B Block Coding mechanisms used in Fast Ethernet and Gigabit Ethernet. Why are they preferred over Manchester? (10 Marks)

• **Manchester Limitation:** Manchester encoding has a baud rate efficiency of only 50% ($S = 2N$), requiring a massive 200 MHz signaling rate for 100 Mbps Fast Ethernet.
• **4B/5B Block Coding Solution:** Takes 4-bit data nibbles and maps them into 5-bit symbols from a lookup table. The 5-bit codes are chosen such that they contain **at most one leading zero and at most two trailing zeros**. When transmitted using NRZ-I, this guarantees at least one transition every 3 bits for continuous receiver clock recovery with only $\mathbf{25\%}$ baud overhead (125 Mbaud for 100 Mbps)!
• **8B/10B Coding (Gigabit Ethernet):** Maps 8-bit bytes into 10-bit symbols, balancing the Running Disparity (DC balance) and providing robust error detection for fiber optic and copper links.

Q9. Differentiate between Frequency-Division Multiplexing (FDM), Time-Division Multiplexing (TDM), and Wavelength-Division Multiplexing (WDM) across signals, media, and multiplexing hardware. (8 Marks)

• **FDM:** Analog multiplexing; divides total channel frequency band into discrete sub-channels with Guard Bands; requires analog bandpass filters (AM/FM Radio, Cable TV).
• **TDM:** Digital multiplexing; interleaves discrete time slices from multiple bitstreams into round-robin frames; requires digital framing logic and buffers (T1/E1, SONET).
• **WDM:** Optical multiplexing; combines multiple optical laser beams of different wavelengths (λ) onto a single optical fiber strand; requires optical prisms and diffraction gratings (DWDM telecommunications).

Q10. Explain Spread Spectrum techniques: Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS). (10 Marks)

Spread Spectrum expands the bandwidth of a narrowband message signal across a wide frequency band using a pseudo-random noise (PN) sequence to provide military-grade anti-jamming and multi-user CDMA capability:
• **DSSS (Direct Sequence Spread Spectrum):** Each data bit is multiplied ($\mathbf{XOR}$) by a high-rate n-bit pseudo-random **Chipping Sequence** (e.g. 11-bit Barker code `10110111000` in IEEE 802.11b Wi-Fi). Even if narrowband interference corrupts part of the band, the receiver cross-correlates with the known chipping code to recover the exact data with high processing gain ($G_p = 10\log_{10}(B_{\text{ss}}/B_{\text{data}})$).
• **FHSS (Frequency Hopping Spread Spectrum):** The carrier frequency rapidly hops across a pseudo-random sequence of frequency channels (e.g. 79 channels in Bluetooth hopping 1600 times per second!). Eavesdroppers and narrow jammers cannot intercept or jam the transmission without knowing the exact hopping seed sequence!

Topic 19.8: Advanced Digital Signal Processing & Orthogonal Frequency Division Multiplexing (OFDM)

Modern broadband wireless and wireline standards (Wi-Fi 6/7, 4G LTE, 5G NR, DSL, DVB-T2) replace single-carrier modulation with **Orthogonal Frequency Division Multiplexing (OFDM)**.

Figure 2.2: Orthogonal Frequency Division Multiplexing (OFDM) Baseband Signal Generation

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> **The Orthogonality Condition in OFDM:** Subcarrier frequencies $f_k = k\Delta f = \frac{k}{T_{\text{sym}}}$ are spaced at exact intervals $\Delta f = \frac{1}{T_{\text{sym}}}$, guaranteeing zero mutual inter-carrier interference (ICI):
>
> $$\mathbf{\frac{1}{T_{\text{sym}}} \int_0^{T_{\text{sym}}} \cos(2\pi f_i t) \cos(2\pi f_j t) \, dt} = \begin{cases} 0 & i \neq j \\ \frac{1}{2} & i = j \end{cases}$$

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